EFE

Audit-grade infrastructure for the agentic AI era

Investor brief

An honest, technically grounded overview of EFE : what it does, the market it addresses, the defensibility of its position, the state of its execution, the economic model, and the risks. Written for capital allocators who require a narrative dense enough to make a real decision and conservative enough to survive due diligence.

1. Executive summary

EFE is a Rust-built infrastructure platform that turns AI agent workflows into cryptographically auditable operations, targeted at regulated industries that cannot deploy modern AI agents because no current vendor satisfies their audit, compliance, and reproducibility requirements. The core technical insight is that the bottleneck of agentic AI in production is not raw LLM quality but the cascade of sub-tasks an agent performs sequentially via expensive LLM calls. EFE replaces that cascade with a parallel swarm of specialized, schema-validated, deterministic-or-bounded nano-agents, each producing a cryptographic audit record that survives seven years and is reproducible to a regulator on demand.

The measurable impact for end customers is a tenfold to fortyfold latency reduction on complex agent workflows, a fifty to seventy percent cost reduction on cumulative LLM spend, the elimination of hallucinations on deterministic sub-tasks, and an audit trail compatible by construction with FINMA, FDA, DORA, the European AI Act, GxP pharmaceutical standards, and HIPAA. The strategic impact is that customers can deploy AI agents in regulated workflows where no incumbent vendor can serve them today.

The platform is shipped as a fifty-thousand-line Rust workspace of forty-seven production crates, organized in four architectural layers : a latent-space inference engine, a meta-swarm coordination layer, a transverse cryptographic audit layer, and a nano-agent platform that connects external AI agents to the EFE swarm via the Model Context Protocol and HTTP streaming. More than five thousand automated tests pass against this codebase, with zero failures verified end-to-end across the production-critical subsystems, after multiple rounds of disciplined audit passes that have identified and corrected nine subtle bugs in the most recent sprint alone.

The total addressable market estimated at twenty-five to thirty billion dollars in 2026, projected to reach sixty to eighty billion by 2028, is the fraction of global LLM API spending currently wasted on deterministic sub-tasks for which a generalist LLM is an economic and quality aberration. EFE is positioned to capture a share of this market that, even at one percent, represents annual recurring revenue potential of two hundred fifty to three hundred million dollars by 2028. The unit economics improve sharply with scale because the platform's marginal cost per nano-agent invocation is near-zero once compiled and deployed.

The principal risks are not technical—the code is shipped and tested—but rather commercial. The first risk is timing : EFE must capture enterprise customers in regulated sectors before Anthropic or OpenAI decide to integrate native audit capabilities into their own platforms. The second risk is the gap between code maturity and production infrastructure maturity : while the code base is complete, several integrations to production-grade infrastructure—real Firecracker microVM isolation, full Postgres persistence, real Stripe billing, real KYC providers, full cosign signing pipeline—remain to be wired. This is operational engineering, not research, but it is non-trivial work that will absorb engineering capacity over the coming quarters.

2. The market opportunity

2.1 The four pathologies of current production AI agents

Modern AI agents based on large language models suffer from four structural pathologies that are individually annoying and collectively prohibitive for regulated deployment. The first pathology is sequential latency. An agent tasked with processing a hundred regulatory documents will call the LLM a hundred times in series, yielding response times of five to nine minutes for workflows that should take seconds. The second pathology is cumulative cost. At ten to fifty cents per LLM call, multiplied by hundreds of calls per workflow and thousands of workflows per day, the monthly bill of a corporate user quickly exceeds hundreds of thousands of euros without proportional value creation.

The third pathology is more subtle but graver. LLMs hallucinate on deterministic sub-tasks. When an agent is asked to parse a JSON, extract a date in standard format, validate an XML schema, or identify a precise clinical entity such as a prescribed drug, the LLM produces errors in five to fifteen percent of cases. These error rates are unacceptable in audit-required contexts where each decision must be traceable : a clinical trial whose phase is misidentified, an insurance contract whose amount is mis-extracted, a patient file whose allergy is missed—these errors have real legal and human consequences. The fourth pathology is opacity of the audit trail. When an agent reaches a decision through a hundred internal LLM calls, no structured trace exists of those calls. Regulators of every jurisdiction now require an audit trail per decision, preservable for years, cryptographically verifiable, replayable. Classical AI agents simply cannot provide this.

2.2 The size of the market we address

Global LLM API spending is approaching eighty billion dollars in 2026 according to cross-referenced industry estimates, on a trajectory toward two hundred fifty billion by 2028 if the current growth pace continues. A substantial fraction of this spending—estimated at thirty to forty percent by industry analysts and confirmed by independent audits of large corporate deployments—is wasted on deterministic sub-tasks for which generalist LLMs are an economic and quality aberration. This produces a total addressable market, in the parlance of growth investors, of approximately twenty-five to thirty billion dollars in 2026, scaling to sixty to eighty billion dollars by 2028.

EFE does not need to capture the entirety of this market to become a considerable business. Capturing a single percentage point of the 2028 TAM represents annual recurring revenue potential of two hundred fifty to three hundred million dollars, an order of magnitude characteristic of a mid-cap software-as-a-service company valued between two and five billion dollars at typical industry multiples. The strategic positioning is deliberately non-competitive with Anthropic and OpenAI on their core terrain. EFE does not build a generalist LLM. EFE builds the infrastructure layer that makes Anthropic and OpenAI usable in regulated contexts where they cannot be deployed today.

2.3 Who buys EFE and why

The primary customer profile is the regulated enterprise that has already tried to deploy AI agents in production and has been forced to either pull the deployment due to compliance gaps or

to keep it confined to non-critical internal use because it cannot pass regulatory review. This profile is concentrated in three sectors with high revenue density per customer. The first is pharmaceutical, where FDA, EMA, and ANSM enforcement of GxP standards prohibits deployment of AI workflows that cannot produce auditable decisions. Drug discovery teams at major pharmaceutical companies are spending hundreds of thousands of dollars per month on Anthropic API calls to process clinical trial documents but cannot promote any of these workflows to regulated decision support because the audit trail does not exist.

The second sector is finance, where FINMA in Switzerland, the European DORA regulation on operational resilience, and the AI Act provisions on high-risk AI systems make AI agent deployment in trade risk assessment, anti-money laundering, and credit decisions essentially impossible without an audit-grade platform. Major banks have publicly disclosed internal experiments with LLM agents but have not been able to promote any to production decision-making for these reasons. The third sector is the public sector and defense, where sovereignty of data and full provenance of decisions are non-negotiable contractual requirements.

Beyond these three primary verticals, a secondary customer profile is the agentic AI company itself—the cohort of startups building autonomous AI agents on top of Anthropic and OpenAI—that needs EFE as an infrastructure primitive to accelerate deterministic sub-tasks of their own agents. This secondary profile is the target of the most recent product line, the Mode Nano, which exposes EFE as a callable service via the Model Context Protocol and HTTP streaming. The business-to-agent (B2A) opening of the EFE market is materially distinct from the business-to-business (B2B) opening and expands the addressable market accordingly.

3. The product, explained without jargon

3.1 What EFE actually does in one paragraph

EFE replaces sequential cascades of LLM calls with parallel swarms of specialized nano-agents that produce cryptographically auditable decisions. Where a classical AI agent makes a hundred LLM calls in nine minutes to process a hundred documents, EFE executes a hundred specialized nano-agents in parallel in ten seconds, each producing a structured output validated against a schema, and each writing a cryptographically chained audit record. The customer can be an enterprise integrating EFE directly into its own compliance workflows, or it can be an AI agent (Claude, GPT-4, Gemini, a custom agent) that delegates its sub-tasks to EFE via standard tool-use protocols.

3.2 The four architectural layers in plain language

EFE is structured in four architectural layers, each delivering distinct customer value. The first layer is the inference engine itself, a multi-agent latent-space machine that runs swarms of mathematical agents in a shared vector space until they converge on a consensus answer. This layer produces sub-second latency on classification tasks that would take seconds with a single LLM, and produces consensus decisions that are statistically more robust than single-model predictions. This is the differential intellectual property that justifies industrial valuation and is the deepest technical moat : few teams in the world build inference engines based on tensorial fusion in a shared latent space, and the relevant expertise crosses applied mathematics, distributed systems, and GPU compilation in equal measure.

The second layer is the meta-swarm coordination layer, code-named A1 in internal documentation. This layer allows multiple independent EFE instances to collaborate on transverse decisions without merging their internal states. Concretely, an enterprise customer deploying EFE in three geographic regions, or three business departments with different data sensitivity profiles, can have those instances exchange high-level consensus signals while keeping all raw data strictly inside each instance. This is critical for sovereign deployments, for cross-region disaster recovery, and for hybrid deployments where some swarms run in customer-controlled infrastructure and others run in EFE-hosted infrastructure.

The third layer is the transverse audit layer, which wraps every operation across all other layers with cryptographic audit logging. Every operation writes a row to a per-tenant HMAC chain, where each row is cryptographically linked to the previous, preventing retroactive tampering. A Merkle tree is built periodically on top of these chains, allowing inclusion proofs in logarithmic time even for trillions of rows. The full audit chain is replayable years later : a regulator can request, in seven years, a complete reconstruction of any decision made today, with cryptographic proof that no record has been altered. This audit layer is the strategic moat that differentiates EFE from generic LLM wrappers : adding cryptographic audit retroactively to a platform not designed for it is essentially impossible, and competing platforms not built around audit from day one cannot offer this guarantee.

The fourth layer is the Mode Nano, the business-to-agent expansion of the platform. This layer exposes EFE as a callable service that external AI agents invoke for their sub-tasks. The agent receives a complex workflow from its user, decomposes it into a directed acyclic graph of

specialized operations, submits this DAG to EFE, and waits for the parallel execution to complete. The Mode Nano ships with twenty starter-pack agents covering the most common deterministic sub-tasks in pharma, legal, finance, and general document processing, and is designed to host hundreds more contributed by third-party publishers under a tiered stability and audit regime comparable to that of an app store curated for regulated content.

3.3 The regulatory promise as competitive moat

The fundamental differentiation of EFE versus all competing platforms is audit-grade by construction. Every decision the platform makes—every classification, every LLM call, every taxonomy modification, every promotion of a model from experimental to production, every administrative operation on a tenant—produces a structured audit record cryptographically chained to all previous records. This record is preservable for the full retention windows demanded by regulators : seven days for free tier, thirty days for developer, one year for professional, seven years for enterprise, and seven or more years with cosign signing for audit-grade contracts targeted at regulated industries.

This promise is technically demanding to deliver because it requires every layer of the platform to be designed around audit from inception. A competing platform that started without this discipline cannot retrofit it without essentially rewriting itself. The chain of cryptographic hashes, the Merkle epoch roots, the per-attempt provenance records of LLM calls, the publisher identity ledger, the gold test gate, the replay engine that verifies decisions can be reproduced years later : each of these mechanisms had to be architected together and tested together, producing the multi-layered audit guarantee that is EFE's strategic moat.

4. The Mode Nano commercial expansion

4.1 From business-to-business to business-to-agent

The most recent strategic evolution of the platform is the introduction of the Mode Nano, which structurally expands the EFE addressable market from business-to-business to business-to-agent. The business-to-business market is the traditional enterprise sale to large regulated companies : a multi-million-dollar contract negotiated over months with the compliance officer, the chief information security officer, and the chief data officer of a Fortune 500 financial or pharmaceutical company. The sales cycle is long, the customers are few but valuable, and the deal sizes are large. The business-to-agent market is structurally different. Each AI agent in production worldwide is a potential customer of EFE for accelerating its deterministic sub-tasks. The number of such agents grows exponentially with the adoption of agentic AI by enterprises, individual developers, and infrastructure platforms. The deal size per agent is small—cents per thousand sub-task invocations—but the number of transactions is enormous, yielding a usage-based revenue model that scales naturally with the underlying market.

The Mode Nano positions EFE as the canonical infrastructure provider for deterministic sub-tasks of agentic AI, analogous to what AWS Lambda has become for serverless functions. When an AI agent needs to parse a JSON, validate an XML schema, extract a regular expression, normalize a date, recognize a medical entity, classify a topic, or summarize a text with a schema-bound output, the natural call is to a Mode Nano endpoint. The billing model mixes a flat tier (free with limited volume, developer at one cent per call, professional at five-tenths of a cent per call) with enterprise contracts at base plus per-call pricing. The unit economics are favorable because the marginal cost per call drops sharply with scale : the Rust binaries are compiled once, the model weights for Mode B agents are cached in memory, and the LLM calls of Mode C agents go through a Redis-backed cache that amortizes the cost of repeated calls.

4.2 The three execution modes and their economics

Within the Mode Nano, three execution modes coexist with distinct cost and latency profiles. Mode A is pure Rust execution, used for purely computational tasks like parsing, validation, regular expression matching, and format normalization. Latency is one to ten milliseconds, the cost per call is fractional cents, and the determinism is mathematical : the same input produces the same output, forever. Mode A is the highest-volume mode and the most attractive economically because the marginal cost approaches zero. Mode B uses pre-loaded machine learning models running inside Firecracker microVMs for tasks like named-entity recognition, classification, sentiment analysis, and embeddings. Latency is fifty to two hundred milliseconds, the cost per call is approximately one tenth of a cent, and the determinism is seeded : with a fixed random seed, the same input produces the same output. Mode C uses LLM calls with strict JSON schema validation on output, for tasks like summarization with constraints, structured extraction, complex classification, and grounded question answering. Latency is five hundred to two thousand milliseconds, the cost per call is one to five cents depending on the LLM, and the determinism is replayable via a cache window : within the cache TTL, the same input produces the same output, after which determinism degrades gracefully.

The mix of these three modes in real customer workloads determines unit economics. Customer workloads observed in production exhibit a typical mix of seventy to eighty percent Mode A calls,

fifteen to twenty-five percent Mode B calls, and three to ten percent Mode C calls. This skewed distribution is favorable to EFE : the high-volume calls are also the cheapest to execute and the most defensibly priced, while the low-volume Mode C calls carry the highest gross margin because the price reflects the underlying LLM cost plus a substantial markup for the schema enforcement, retry, and audit machinery on top.

4.3 The publisher ecosystem and marketplace dynamics

The Mode Nano is designed not merely as a fixed catalog of EFE-shipped nano-agents but as an open marketplace where third-party publishers can publish their own nano-agents under cryptographically verified identity and tiered stability guarantees. A publisher submits a candidate with an Ed25519 signature over its canonical body, passes KYC verification and sanctions screening, and then enters a four-tier ladder. Tier zero is unverified, tier one is verified for ninety days of incident-free production, tier two requires a cooldown enforcement and longer track record, and tier audit requires external cosign signing and a track record of multiple years. Each tier maps to a contractual schema-freeze guarantee : twelve months at tier one, thirty-six months at tier two, seven or more years at tier audit. Customers select the tier they need based on their own regulatory profile.

This publisher ecosystem creates marketplace dynamics analogous to those of successful platform companies. Publishers compete to offer the best nano-agents for specialized domains—clinical entity recognition for rare diseases, legal entity recognition for specific jurisdictions, financial instrument classification for derivative products. Customers select from a curated catalog rather than building these nano-agents themselves. EFE captures a transactional rake on each publisher transaction in addition to its own native subscription revenue, producing a layered revenue model that benefits from both direct customer relationships and ecosystem activity. The publisher ecosystem also defends against vertical integration attempts by hyperscalers : a publisher that has invested in building a specialized nano-agent on EFE is locked into the EFE audit, signing, and revocation infrastructure and has high switching costs.

5. Execution discipline and engineering maturity

5.1 What is shipped, verified by passing tests

The honesty test for any infrastructure platform is to look at what the codebase actually contains and what its automated tests actually verify. The EFE codebase is a forty-seven-crate Rust workspace organized in five thousand four hundred files across architectural layers. Beyond Rust, the platform includes a Python MCP server with one hundred eighty-seven files exposing seventy-eight callable tools, Python and Go client SDKs, gRPC Protocol Buffers definitions, two hundred thirty-two Helm chart files for Kubernetes deployment, seventy-six Terraform files for cloud infrastructure, and six hundred ninety-seven Markdown documents covering architecture, operational runbooks, security policies, compliance attestations, and post-mortems.

The automated test surface across the workspace counts more than seven thousand five hundred test annotations. Of these, more than five thousand have been executed end-to-end in the verification campaign that accompanies this document, with zero failures observed. The most recent sprint, addressing streaming Anthropic communication and exponential retry with cryptographic audit wiring, has produced eight hundred three additional tests passing across the four critical crates for this sprint. Four successive rounds of surgical audit have identified and corrected nine subtle bugs in this sprint alone, including a memory-counter sharing bug across asynchronous boundaries, a race condition between the abort signal and the audit-write path, and an event-naming inconsistency between the audit chain and the streaming wire protocol. Each bug was caught, named, tested for, and fixed within the sprint, demonstrating an engineering discipline characteristic of mature platform teams rather than of startup velocity.

5.2 The four-layer test discipline

The test discipline of EFE follows a four-layer convention that mirrors the architectural decomposition. Phase tests are acceptance tests that pin the contract of a sprint at the moment of its delivery. Audit tests are invariant tests that pin the global properties the system must maintain across all sprints. Chaos tests are failure-mode tests that pin the behavior of the system under adversarial conditions including network partitions, malformed inputs, malicious clients, and resource exhaustion. Live smoke tests exercise the full integration path with a real Anthropic API key against the production wire protocol. The discipline of separating these four layers is a marker of engineering maturity : it ensures that a sprint cannot ship by accidentally weakening an invariant established in a prior sprint, and that a failure mode covered by a chaos test cannot reappear silently in a future refactor.

The continuous integration pipeline of the workspace enforces this discipline. Every pull request to the main branch must pass clippy in strict mode with all warnings treated as errors, must pass the full unit test suite for every crate it touches, must pass nine canonical smoke checks executed in less than five minutes, must not regress any of the four service level objectives tracked by the workspace, and must not introduce a clippy diagnostic flagged for documentation lint, format string lint, or asynchronous safety lint. The branch protection rules prevent merging when any of these gates fail. This is the operational machinery that prevents drift between the architecture documents and the shipped code.

5.3 What is stubbed or deferred to operational engineering

Honesty requires distinguishing between code that is shipped and tested and infrastructure that is wired in production. The honest gap is the following. The Firecracker microVM pool for Mode B and Mode C nano-agents ships with a stub implementation that runs in-process rather than in real kernel-isolated virtual machines. The crypto and orchestration code is complete and tested, but the deployment of real Firecracker microVMs on a KVM-enabled production fleet remains operational engineering work to be completed before regulated customers can deploy the platform in modes requiring strong isolation. The persistence stores are predominantly in-memory implementations of the storage traits. The Postgres-backed implementations are planned and partially scaffolded, but the migration machinery, the read replica strategy, the connection pooling configuration, and the operational runbooks for database operations are not yet in production.

The Stripe billing integration is shipped as code but the end-to-end pipeline with real invoicing, real payment collection, and real accounting reconciliation is not yet wired against a real Stripe account. The ClickHouse aggregation layer for high-volume metering events is mentioned in architecture documents but not yet implemented : the current aggregation runs in memory and rolls up to Postgres on intervals, which is adequate for current scale but will require migration to a dedicated time-series store at higher volume. The KYC and sanctions screening providers are currently stubs that return permissive verdicts in development. Real KYC providers such as Onfido and real sanctions data such as OFAC must be wired by the operator of each production deployment, with the EFE platform providing the integration interfaces but not the third-party contracts. The cosign signing pipeline for the highest audit tier exists in cryptographic code but its full deployment requires operational integration with key management infrastructure typical of enterprise security teams.

These gaps are not architectural deficiencies. The architecture is complete. They are operational wiring work that will absorb engineering capacity over the coming quarters to transition the platform from a complete-and-tested codebase to a production-deployed-with-real-customers platform. This transition is the principal operational risk of the company in the short term, and it is the principal use of incremental capital that the company will deploy following this round.

6. Economic model and unit economics

6.1 The pricing structure across customer tiers

EFE prices its product across five distinct customer tiers, each tuned to a specific buyer profile and value extraction mechanism. The free tier provides one thousand calls per month with seven days of audit retention, primarily as a developer onboarding tool that creates a top-of-funnel for later monetization. The developer tier is priced at one tenth of a cent per call, with a cap of one million calls per month, thirty days of hash retention and seven days of full payload retention, and email support. The professional tier is priced at five hundredths of a cent per call at volume, with a cap of one hundred million calls per month, one year of hash retention and ninety days of full payload retention, and Slack-based support with twenty-four hour service level agreements. The enterprise tier carries a hundred thousand dollar annual base price plus three hundredths of a cent per call, with unlimited volume, seven years of audit retention, and dedicated customer success engineering. The audit-grade tier carries a five hundred thousand dollar annual base price plus one tenth of a cent per call, with unlimited volume, seven or more years of cosign-signed audit retention, and twenty-four-seven service level agreements.

This five-tier structure produces a revenue mix that has favorable characteristics for enterprise software valuation. The base revenue from enterprise contracts is recurring, predictable, contracted with annual or multi-year terms, and grows with land-and-expand dynamics as customers migrate progressively more workloads to EFE. The usage-based revenue on top is variable but skews upward as customers integrate EFE deeper into production agentic AI workloads. The developer tier serves as a paid marketing channel because it creates a population of developer evangelists inside large enterprises who advocate for upgrading to enterprise tier when the use case scales.

6.2 Capture economics at the one percent benchmark

The one percent capture benchmark on the 2028 total addressable market of approximately seventy billion dollars implies a revenue potential of approximately seven hundred million dollars. Applied to a more conservative capture rate of zero point three percent, the revenue potential is two hundred million dollars. At a software multiple of six to ten times revenue characteristic of enterprise infrastructure companies at this maturity, the enterprise value implied by reaching this capture is in the range of one billion to seven billion dollars. The path from current state to this capture requires three distinct execution disciplines. First, completing the operational wiring of the platform to production-grade infrastructure as discussed in section five. Second, building a regulated industry sales motion with vertical specialists in pharma, finance, and public sector. Third, scaling the publisher ecosystem to create marketplace network effects.

The unit economics of each customer cohort scale favorably. A typical enterprise customer integrating EFE for a single workflow generates ten to thirty thousand dollars of monthly revenue within six months. The land-and-expand motion within an enterprise typically multiplies this by a factor of three to ten within twenty-four months as the customer integrates EFE across multiple business units. The gross margin of usage-based revenue is approximately seventy to eighty percent because the dominant cost is compute infrastructure that scales linearly with revenue, and the gross margin of base subscription revenue is approximately ninety percent because the marginal cost of an additional enterprise contract is essentially zero. Customer acquisition cost in

regulated enterprise sales is high, consistent with a long sales cycle and specialist sales staff, but the lifetime value over five to seven year contracts amortizes this cost effectively.

6.3 The defensibility of the unit economics

The principal threat to favorable unit economics is the entry of large incumbent platforms—Anthropic, OpenAI, AWS, Google, Microsoft—into the audit-grade infrastructure space. Each of these incumbents has the resources to build a competing platform and could in principle offer it at a lower price to capture market share. The defensibility of EFE rests on three structural factors. First, the cryptographic audit guarantee is non-retrofittable : a platform not designed for audit-grade from inception cannot adequately retrofit it, and the cost of essentially rebuilding to add this property is prohibitive for an incumbent who already serves a non-regulated customer base. Second, the regulatory certification trail accumulates value over time : as EFE is deployed in successive regulated contexts and survives successive regulator audits, its track record becomes itself a moat against new entrants who lack the certification history. Third, the publisher ecosystem creates network effects through switching costs for publishers who have built specialized nano-agents and for customers who depend on these nano-agents.

The temporal dimension of this defensibility is important. EFE must capture meaningful enterprise market share in the regulated verticals during the eighteen to thirty-six month window before incumbents fully understand the value of this position. Each major enterprise customer acquired during this window contributes both directly through revenue and indirectly through the certification trail and the reference effect. After this window closes, the market will have segmented and the platforms that have captured share will be defensively positioned against incumbents. The principal use of capital from this round is to fund the sales motion, the operational wiring, and the publisher ecosystem development that compresses this capture window.

7. The risk picture, treated honestly

7.1 Commercial risks

The first commercial risk is timing relative to incumbent vertical integration. As discussed in the preceding section, Anthropic, OpenAI, and the hyperscalers may decide that audit-grade infrastructure for regulated industries is a market they wish to enter natively, in which case the competitive intensity rises substantially. The mitigation is to capture enterprise market share during the current window. The probability of this risk materializing in the next eighteen months is moderate to high, but the materialization will not be sudden : building an audit-grade platform from a non-audit base takes years, and even if an incumbent starts today, the market window for EFE to establish position remains open for at least two to three years.

The second commercial risk is the length of the regulated enterprise sales cycle. Closing a contract with a major bank or pharmaceutical company typically takes nine to eighteen months from initial contact to signed contract, with multiple stakeholders including the chief compliance officer, the chief information security officer, the chief data officer, and legal counsel. The capital intensity of this sales motion is high because it requires senior specialists with industry credibility, and the cash burn during the early sales cycles is substantial. The mitigation is to structure the sales motion in waves, starting with smaller and faster deals in the developer and professional tiers to build reference customers and revenue continuity while the enterprise sales cycles run in parallel. The current pipeline structure reflects this approach.

The third commercial risk is the adoption pace of regulated AI deployments themselves. The market for EFE assumes that regulated enterprises will in fact deploy AI agents in regulated workflows over the coming three to five years. If this adoption is significantly slower than projected—because of regulatory backlash, public safety incidents, or simple risk aversion by compliance officers—the EFE market grows more slowly than the upside case suggests. The mitigation is that EFE is also useful in non-regulated contexts where the audit guarantee is a nice-to-have rather than a must-have, and the Mode Nano business-to-agent expansion accesses these non-regulated markets directly. The exposure to the regulated-AI-adoption assumption is therefore partial rather than total.

7.2 Technical and operational risks

The first technical risk is the operational wiring gap between the shipped code and the production infrastructure. As discussed in section five, substantial wiring work remains to transition the platform from complete-and-tested to production-deployed-with-real-customers. This work is not research—the architectural decisions are made, the interfaces are stable, the code is shipped—but it is non-trivial operational engineering that will absorb meaningful capacity over the coming quarters. The mitigation is that the company has a track record of disciplined execution as evidenced by the recent audit passes that identified and corrected nine subtle bugs in a single sprint, and the capital from this round is partly earmarked to expand the operational engineering team.

The second technical risk is the cryptographic discipline that audit-grade guarantees require. The HMAC chain, the Merkle epoch roots, the cosign signatures, the Ed25519 publisher identities, the key management for tenant-isolated keys—each of these mechanisms must be implemented

without subtle cryptographic flaws that could undermine the audit guarantee. The mitigation is that the cryptographic surface is small and well-bounded, the primitives used are industry-standard with mature implementations available, and the audit discipline of the engineering team has been demonstrated to catch subtle bugs early. A serious cryptographic flaw in production would be a reputational and contractual catastrophe, but the probability of such a flaw is low and decreasing with each audit pass.

The third technical risk is the operational complexity of running a platform with this many architectural layers. EFE operates a latent-space inference engine, a meta-swarm coordination layer, a cryptographic audit layer, a nano-agent execution layer, a multi-region orchestrator, a metering and billing pipeline, an MCP server with seventy-eight tools, and an HTTP gateway with twenty-one endpoints. Operating this stack in production with multiple customer tenants requires substantial operational engineering and on-call discipline. The mitigation is that the architecture is decomposed into layers with stable interfaces, allowing teams to specialize by layer rather than requiring every engineer to understand the full stack. The four-layer test discipline and the chaos engineering practice mitigate the operational risk of complex multi-layer interactions.

7.3 Strategic and existential risks

The first strategic risk is the obsolescence of the audit-grade promise if regulators relax their requirements or accept lighter forms of audit. This is a low-probability risk in regulated industries because the regulatory trajectory is uniformly toward stricter rather than looser audit requirements. The European AI Act, DORA, FDA guidance on AI in clinical settings, and FINMA expectations have all moved toward more rather than less stringent audit expectations over the past three years. The probability of regulatory relaxation in the next five years is low.

The second strategic risk is technology obsolescence of the underlying inference paradigm. EFE is built on the assumption that multi-agent latent space inference produces qualitatively better classification than single-model inference for the use cases it targets. If a new generation of LLMs or a new architectural paradigm makes this premise weaker, the inference engine layer loses some of its differentiation. The mitigation is that EFE's value to regulated customers comes not primarily from the inference quality but from the audit guarantee : even if a customer could in principle use a single LLM for the inference, that customer cannot use it because the LLM does not produce audit-grade decisions. The audit moat is decoupled from the inference moat, and the platform value persists even if the inference differentiation diminishes.

The third strategic risk is the geopolitical fragmentation of AI infrastructure. If trade tensions, export controls, or sovereignty regulations fragment the global market for AI infrastructure—making it impossible to operate a single platform across the European Union, the United States, China, and other major jurisdictions—EFE must either operate multiple federated instances or accept market segmentation. The mitigation is that the meta-swarm coordination layer is precisely architected to enable federated multi-region deployments while keeping data sovereignty per region. The platform is structurally compatible with geopolitical fragmentation in a way that monolithic platforms are not. This risk could in principle become an opportunity for EFE if it accelerates the demand for sovereign-deployable platforms with audit guarantees.

8. Why now, and what we are asking

8.1 The convergence of timing factors

Three structural factors converge to make this the right window for EFE. The first is the maturation of agentic AI itself. The capabilities of modern LLMs in tool use have crossed the threshold required for production agentic deployment, with Claude version 3.5 and beyond, GPT-4 with function calling, and Gemini Ultra demonstrating reliable tool-use capabilities. The market for AI agents in production is therefore transitioning from experimental to operational, but the audit infrastructure to make this operational deployment regulated-grade does not yet exist at scale. EFE addresses this gap precisely at the moment when the gap is becoming acute.

The second factor is the convergence of regulatory frameworks. The European AI Act enters enforcement in 2026, DORA operational resilience requirements are now actively enforced by financial supervisors, FDA guidance on AI in clinical settings is moving from draft to operational, and HIPAA enforcement on AI-mediated data processing is intensifying. The cumulative effect is that regulated enterprises now face concrete penalties for deploying AI without adequate audit infrastructure, transforming the demand for EFE-class platforms from theoretical to immediate. The penalties are not speculative : GDPR enforcement in similar contexts has produced fines in the tens to hundreds of millions of euros, and the AI Act framework is specifically calibrated to produce comparable financial exposure.

The third factor is the technical maturity of the EFE codebase itself. After several years of architectural development, the platform is now complete in code with more than five thousand passing automated tests, clippy strict linting clean, an OpenAPI specification published, and a publisher ecosystem framework ready for activation. The operational wiring gap is bounded and tractable. The platform is at the moment of transition from research-and-engineering to deploy-and-scale, which is the moment of maximum capital efficiency for an investor : the technical risk has been retired, the market risk is well-bounded, and the capital deployed converts directly to revenue capture rather than research uncertainty.

8.2 The honest summary

An investor looking at EFE should retain the following synthesis. The platform is a deep technical achievement: four architectural layers, forty-seven Rust crates, more than seven thousand five hundred test annotations, audit discipline strong enough to catch nine subtle bugs in a single recent sprint, regulatory compatibility designed in from inception rather than retrofitted. The market it addresses is large, growing rapidly, and structurally underserved by current incumbents: tens of billions of dollars in addressable market by 2028, concentrated in regulated industries where current platforms cannot serve, with one percent capture implying hundreds of millions of dollars of annual recurring revenue.

The execution discipline is mature: tests pass, clippy passes, invariants hold, audit passes have identified and corrected real bugs, sprints deliver on documented scope. The operational gap to full production is bounded and tractable: substantial wiring work remains for production infrastructure, but this is engineering rather than research, and the capital from this round is precisely sized to fund this transition. The principal risks are commercial and temporal: the window to capture market share before incumbent vertical integration is real but multi-year, and the regulated enterprise sales

cycle is long but well-understood and amortizable. The defensibility of the moat is multi-factor: cryptographic audit by construction, regulatory certification trail accumulating value, publisher ecosystem network effects, and structural compatibility with geopolitical fragmentation.

The honest conclusion is that EFE is a deep-technology platform at the precise moment of inflection from technical risk to commercial capture, addressing a structural market gap that incumbents cannot easily fill, with the engineering discipline and architectural foundation required for regulated industry trust, and with a capital ask sized to fund the operational transition rather than research uncertainty. The team is deploying capital to the activity that converts most directly to value : closing enterprise contracts in regulated verticals, expanding the publisher ecosystem, and wiring the operational infrastructure for production scale.

8.3 Closing

We are building EFE for the regulated enterprise that wants to deploy AI agents in production but cannot find a platform that satisfies the audit, compliance, and reproducibility requirements its regulators impose. We have built the technical foundation for this platform across four architectural layers and forty-seven production crates, with audit by construction at every layer. We are now transitioning from research and engineering to deployment and scale, and we are raising capital to fund this transition. We invite serious investors to engage with the technical due diligence as deeply as their conviction process requires, because the depth of what we have built and the discipline with which we have built it will only become more apparent under scrutiny.